

## SPECIFIC AIMS

Attention-deficit/hyperactivity disorder (ADHD) is a common (11.4% of United States (US) children 3-17yo and 4.4% of US adults) neurodevelopmental condition characterized by persistent inattention or hyperactivity-impulsivity that impairs daily function. Currently, the best treatments for ADHD are psychostimulant medications. Unfortunately, stimulants are associated with troublesome side effects and risk of diversion for illicit use and remain underutilized and stigmatized. With a total societal excess cost attributable to ADHD estimated at >\$122 billion per year,<sup>1</sup> **there is a pressing need for more targeted approaches with potentially more lasting benefits and fewer side effects**. A key step toward this goal is to gain actionable insight into the therapeutic mechanism of stimulants for translation to more targeted interventions with fewer side effects.

We know that studies report measurable changes in brain physiology in response to stimulant use, information which could potentially guide targeted neuromodulatory interventions, such as transcranial magnetic stimulation (TMS) and fMRI-based neurofeedback<sup>1</sup>. Pharmacological fMRI (phMRI) has provided the best quantitative assessment of these changes, and rigorous prior research with this approach has established that brain networks involving the dorsal anterior cingulate cortex (dACC) and the right inferior frontal gyrus (rIFG) are modulated by pharmacological treatment of ADHD during task-related effort. However, the phMRI literature reports variable effect sizes and directions of signal change in these regions in association with treatment response. These prior studies note that, in addition to known origins of heterogeneity (e.g., methodology, pathophysiology), they are likely also confounded by head motion artifacts. However, the impact of variation in ADHD-related head motion on imaging findings has not been studied, so it remains unclear which published phMRI findings truly represent the mechanism for stimulants' beneficial therapeutic effects, posing a barrier to developing improved therapies.

*The current project asserts that we need to define and correct for the role that head motion plays in imaging studies of ADHD before we can define where measurements of stimulant-response truly correlate with symptom improvement.* The proposed study team has significant experience with identifying patient-level biomarkers in ADHD and developing and implementing advanced motion correction in neuroimaging, recently demonstrating that "slice-based" motion correction captures significantly more head motion than volume-based techniques.<sup>2,3</sup> Our preliminary results demonstrate the importance of better motion correction in patients with ADHD and a clear interaction with stimulant use, prompting the two aims of our proposal:

### **Aim 1: To study the effects of superior motion correction on outcomes of within-subject task phMRI in stimulant-responsive ADHD by comparing standard and advanced (slice-by-slice) motion correction.**

**Objective 1:** To define effects of stimulants in ADHD on the regions most commonly showing task-related changes in activation (IFG, dACC) after improved motion correction. We will compare task-related findings in the IFG and dACC using our recently developed slice-wise motion correction vs. standard volume-based motion correction in high-motion vs. low-motion patients. *We hypothesize that slice-corrected motion correction algorithms will be associated with less response heterogeneity across participants and that low-motion patients will show less response heterogeneity independent of correction approach.*

**Objective 2:** To identify other brain regions that show stimulant treatment-related changes in ADHD after superior motion correction, including regions detected inconsistently in the literature. *We hypothesize that we will be more likely to identify stimulant effects on subcortical regions, which are more prone to false negatives with movement.*

### **Aim 2: To study the effects of superior motion correction on outcomes of within-subject on-/off-stimulant rs-fcMRI (connectivity) by comparing standard and advanced (slice-by-slice) motion correction.**

**Objective 1:** To compare resting-state seed-based connectivity with the rIFG and dACC between slice-wise motion correction vs. standard volume-based motion correction in high-motion vs. low-motion patients. *We hypothesize that there will be differences in network findings between volume- and slice-wise motion correction algorithms, and less heterogeneity in slice-corrected findings and low-motion patients.*

**Objective 2:** To compare functional connectivity network differences on-/off-stimulant using slice-wise vs. volume-wise motion correction for the salience and parietal memory networks, relative to the dorsal attention network. *We hypothesize that, with better motion correction, connectivity will be stronger in the salience and parietal memory networks than in the dorsal attention network.*

**Impact:** In this study, we will leverage phfMRI techniques, an intra-patient design, and state-of-the-art slice-based fMRI motion measurement to identify biomarkers and potential neuroanatomical treatment foci for functional deficits in ADHD. If successful, this project will support development of tailored neurotherapeutic interventions that align with NIH's mission to advance personalized medicine and improve mental health outcomes.

## RESEARCH STRATEGY

### A. Significance

#### ***A1. Importance of the Problem to be Addressed.***

Attention-deficit/hyperactivity disorder (ADHD) is a neurodevelopmental condition characterized by varying combinations of difficulty focusing on tasks, impulsivity, and inability to sit still.<sup>4</sup> ADHD typically begins in childhood, but very frequently persists into adulthood, with 60-90% of those affected in childhood still experiencing symptoms in adulthood.<sup>5,6</sup> The condition is associated with substantial difficulties in school and work performance, and with challenges in overall life management and relationships. Since attention and behavioral control are crucial aspects to nearly every aspect of life, the burden of ADHD is substantial and broad, affecting quality of life of individuals and their families, school/work performance, and ultimately, the economy. The economic burden of ADHD is estimated to be \$122 billion per year in the US.<sup>1</sup> One study showed that costs associated with childhood expenses were more than five times greater for families with children affected by ADHD than for those not impacted by ADHD; this included expenses related to categories such as parental missed work, tutoring, legal fees, and lost items.<sup>7</sup>

There are treatments for ADHD that are well-established for efficacy in a large percentage of those affected, e.g., psychostimulants, which are believed to target dysfunctional striato-cortical circuitry.<sup>8</sup> However, these stimulants are systemic pharmacologic agents, rather than targeted treatment, and—even when very effective treating ADHD symptoms—often come with a host of unwanted side effects, such as suppressed appetite, sleep difficulties, potential abuse/dependence/diversion, and/or worsening of commonly comorbid conditions, such as mood disorders, agitation/irritability/aggression, and anxiety.<sup>9,10</sup> This can lead to new problems and a need for additional medications or medication non-compliance. Thus, it is critical to develop better, more targeted treatments for ADHD. Although the action of stimulant medications at the synapse itself is well established,<sup>11</sup> we currently do not know the precise mechanism(s) by which stimulants exert their clinical and behavioral effects,<sup>12</sup> in part because of inconsistent effects reported across the brain imaging literature.

In the current project, we will aim to move one step closer to identifying the precise mechanisms of stimulants in ADHD by defining and ameliorating a major confound that may be masking true mechanisms of stimulant action, namely, the interdependency of (1) improved ADHD symptoms and (2) reduced head motion in the MRI scanner that is variable across the patient population. Medications that treat hyperactivity are clearly likely to impact overall movement in the scanner, which necessarily includes head movement. Head motion is a major source of artifacts in brain imaging, and yet successful treatment of the hyperactivity component means that many individuals' contrasts of "off vs. on" medication may simply reflect significant artifacts vs. reduced artifacts instead of, or in addition to, the actual mechanisms of action for targeted neurobehavioral symptoms.

Head motion is always a challenge for imaging studies,<sup>13</sup> particularly pediatric studies,<sup>14,15</sup> but when there is a critical relationship between the clinical effect itself and the amount of head motion, as there is in ADHD,<sup>16</sup> the importance and detriment of this factor to interpretation of outcomes is significantly amplified and cannot be overlooked.

#### ***A2. Rigor of Prior Research Supporting the Aims***

##### **Imaging literature and importance to treatment thus far**

phMRI was developed as an approach to interrogating brain mechanisms underlying the effects of neurological or psychiatric treatments. That said, many treatments have been developed and implemented without knowing the precise mechanisms by which they exert therapeutic effects, including psychostimulants.<sup>17</sup> phMRI helps to define the regions of the brain that are critical for relieving symptoms, potentially without causing unwanted effects, to advance or optimize treatment dosing or protocols, or to achieve clinical efficacy with other modalities that are less invasive and less likely to induce side effects than systemic medications. While phMRI is theoretically an excellent tool, there are critical confounds that make interpretation of findings challenging. Among these are neural correlates of side effects, and neural correlates of behavioral effects that lead to different quality of images before and after treatment. In the case of ADHD, the condition itself is frequently associated with hyperactivity, which often leads to greater head motion during MRI scanning sessions when untreated, compared to after effective treatment.<sup>16,18</sup> As a result, treatment itself reduces artifacts that produce potential false positives or negatives that have nothing to do with changes in brain activation underlying the mechanism of action. While this is frequently mentioned as a comment in the limitations sections of ADHD imaging publications, it has not been addressed directly to determine how much it impacts findings or what to do about this significant challenge. **In the proposed studies, we will tackle this issue head-on and aim to**

## **identify just how significant a problem this is, and test one way in which to correct for it: Improved motion correction algorithms.**

**Aim 1:** There is a growing body of pHMRI studies of varying sample sizes that have aimed to evaluate the mechanisms of stimulants in pediatric ADHD patients.<sup>19-28</sup> pHMRI compares task fMRI “off versus on” medications. The two regions most commonly showing altered task-related activity in response to stimulants are the IFG (most often the right hemisphere) and bilateral dACC. However, not every study detects changes in each of these regions, and some studies show increased task-related function in these two regions with stimulant, while others show reduced task-related function with stimulant in these same regions. For example, Rubia and colleagues showed increased activation of dACC with stimulant, while several other studies, including studies by the same group, found reduced dACC activity.<sup>19, 21</sup>

Other regions identified sporadically with stimulant pHMRI include putamen and insula, as well as a range of other cortical regions and cerebellum,<sup>19</sup> but these regions are not detected consistently, which is concerning without an obvious reason for these differences. While there may be multiple reasons for study outcome heterogeneity, including underlying differences in disorder etiology, duration of previous treatment,<sup>29</sup> and variations in task/study methodology, in the current project, we aim to address a likely reason for heterogeneity that is commonly cited that is *not* biological, but artifactual: differences in motion artifacts due to the efficacy of stimulants in reducing hyperactivity with effective treatment. None of the critical phenotype–endotype relationships (i.e., relating to side effects, method, true response result, etc.) can be disentangled without first removing non-biological confounds from recorded signals.

**Aim 2:** Similarly, there is existing literature on changes in resting state fMRI (rs-fMRI) connectivity on- vs. off-stimulant in pediatric ADHD, using pharmacologic resting state fMRI (phrs-fMRI).<sup>30, 31</sup> Given the network nature of this approach, the literature and findings are discussed differently from task pHMRI and thus must be examined independently. These studies most commonly support changes in the default and attention/cognitive control networks associated with stimulants. However, these studies have varied a great deal in sample size, and recent, larger-scale studies have identified changes in regions associated with arousal and reward rather than the other networks.<sup>32</sup> Since head motion influences measures of short-range connectivity due to correlated motion, any effects of movement will be compounded as more regions are added;<sup>14, 18, 33, 34</sup> thus, the effects on functional connectivity may be even more significant than on task-related activation.<sup>35</sup> Since resting state fMRI imaging also, by nature, is more passive than task-related fMRI, there is also a likelihood that participant boredom leads to increased difficulty remaining immobile in individuals with ADHD.<sup>36</sup>

A recent large-scale cross-sectional analysis of stimulant use in the Adolescent Brain Cognitive Development (ABCD) study<sup>37</sup> identified consistent resting-state functional connectivity differences in the somatomotor and salience networks—distinct from those reported in the task-based pHMRI literature;<sup>32</sup> this emphasizes not only the potential for movement confounds leading to across-study inconsistencies, but also the importance of studying both task pHMRI and rs-fMRI; these data are complementary, and each is needed to understand the complete ADHD landscape. However, we cannot disentangle which differences are real vs. result from differential motion effects, without first better characterizing and accounting for the motion effects themselves. This is particularly true since motion affects task- and resting-state analyses in different ways.<sup>34</sup> For example, it has been shown that, in general, imaging research participants move more during resting state than during task-related imaging.<sup>35</sup> However, given the difficulty focusing on tasks for individuals with ADHD, it is unknown how this applies to this population, leaving the possibility that they may move even more during task runs than during rest or vice versa. In addition, motor responses that are part of tasks may also affect head movement, but for mechanical reasons, while during resting state fMRI, movement is more likely to be due to restlessness, leading to different movement profiles.

### **Gaps in the literature and potential reasons for these gaps**

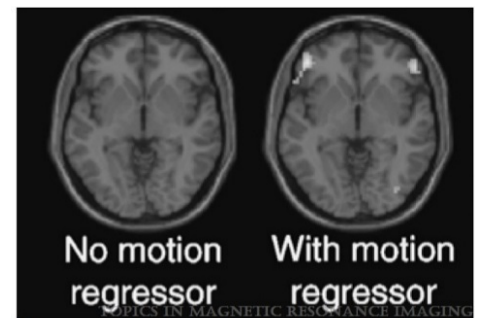
While all of these studies employed the current standards of motion correction,<sup>38</sup> volume-wise correction, and potentially motion scrubbing of contaminated timepoints, this standard method does not account for within-volume movement, only average between-volume movement. Each volume is acquired as a sequence of slices over the course of 1-3 seconds and then “timing corrected” or just assumed to represent a single moment in time, but this process breaks down if patients move within volume, since all slices are not actually acquired simultaneously in a single moment in time. Participants are just as likely to move within-volume as between-volume. If participants move during a volume acquisition, slices are acquired from a different region of the brain than intended; thus, data from some brain regions are missing and others are double-acquired, but you don’t know that unless you register with a slice-based algorithm. To counteract the effects of head motion and its covariance with treatment efficacy, improved motion correction methods are essential.

Our team recently developed two important technologies to address this issue: 1) within-slice motion characterization and 2) fMRI sparse-matrix completion,<sup>2,3</sup> potentially reducing motion artifacts by an order of magnitude proportional to the number of slices acquired. *In the current project, we will apply these superior motion correction algorithms to evaluate how much this factor contributes to variability in imaging studies of stimulants in ADHD.* Dr. Cohen (PI) has extensive experience with fMRI and fcMRI analyses, with 20 years of publications identifying functional networks across development, Dr. Warfield (Co-I) leads a world-class laboratory focused on advancing neuroimaging techniques, including post hoc and real-time slice-motion characterization and correction, and Dr. Arnett (Co-I) has many years of experience using cutting-edge electrophysiological methods to study the heterogeneity of neurocognitive profiles in ADHD, including EEG biomarker development.

Dissociating the effect of stimulant use on 1) altering brain activity in function networks to improve attention vs. 2) reducing head motion leading to uneven reduction in signal-to-noise would clarify the development of novel targeted neurotherapeutics, such as TMS and fMRI neurofeedback (fMRI-NF). Not untangling this issue perpetuates the confounding of neuroimaging studies in ADHD, impeding progress in identifying both biomarkers and potential neuroanatomical foci for executive function deficits.

### Limitations of standard motion correction and impact of misalignment on imaging findings

While it seems obvious that head movement would produce artifacts in imaging, it is worth highlighting the precise manner in which head motion impacts images and imaging findings. Imaging relies on alignment of a given region across time and across task/control conditions to yield physiologically meaningful data. There are different ways that movement can affect outcome/findings, and while these can sometimes be obvious when looking at images, other times the impact is indistinguishable from real signal. For example, given the contrast at the border of gray/white areas—both cortically and subcortically—there will be motion-related artifacts if alignment is not precise, since these regions differ substantially in signal intensity, suggesting changes in task-related signal between conditions where there were none (Type I error). In addition, due to their small size, subcortical regions may be particularly susceptible to reduced replicability across studies due to motion artifacts; some critical subcortical regions are so small that even a slight shift of anatomy due to suboptimal motion correction will lead to non-overlap of the entire region across repeated measures studies. This may then lead to type II errors due to loss of signal (e.g., **Fig. 1**).<sup>3, 39, 40</sup>



**Figure 1.** Results of the same statistical analysis with and without the inclusion of motion estimates as a regressor in the statistical model. The subject (a 10-year-old girl) performed a phonological awareness task compared to a visual matching task (statistical map thresholded at  $p < 0.0001$ , uncorrected for multiple comparisons). From Poldrack et al 2002.<sup>39</sup>

### A3. Significance of the Expected Research Contribution

Upon successful completion of the proposed research, we expect our contribution to (1) articulate the degree of cross-study inconsistency that is due to treatment-related motion effects and (2) demonstrate that this can be alleviated by applying superior motion correction algorithms. *This contribution is expected to be significant because there have been no prior attempts to determine the degree to which study heterogeneity results from motion artifacts or to remove these artifacts.* The credible and tangible benefits of this contribution include: 1) providing a methodological framework to reconcile decades of inconsistent fMRI findings in ADHD; 2) identifying accurate, robust neuroanatomical targets for therapeutic intervention that are not confounded by motion; and 3) directly enabling the development of precise, non-pharmacological neuromodulatory treatments (e.g., TMS or fMRI-NF) that align with the NIH's mission to improve mental health outcomes while minimizing systemic side effects.

Our long-term goal is to recapitulate the fMRI signature—and potentially the benefits—of stimulant use in ADHD using neuromodulatory therapy, e.g., real-time fMRI-Neurofeedback or Transcranial Magnetic Stimulation. The overall objective of this application, which is the next step toward attainment of our long-term goal, is to evaluate the potential confound of head motion on our ability to accurately identify the brain regions that must be targeted to achieve efficacious treatment. Specifically, we will evaluate whether slice-corrected motion correction algorithms will be associated with less response heterogeneity across participants and whether low-motion patients will show less response heterogeneity independent of correction approach (Aim 1), and whether there are differences in network findings between volume- and slice-wise motion correction algorithms and less heterogeneity in slice-corrected findings and low-motion patients (Aim 2) in 150 adolescents with

ADHD chronically treated with (dex)methylphenidate. These findings will establish the importance of motion correction precision in this motion-prone population and separate true therapeutic effects from motion artifacts.

## B. Innovation

Although we (and others) have utilized fMRI and phMRI to characterize the dACC and rIFG's role in executive function in ADHD and in response to stimulant use, the research proposed in this application is innovative, in our opinion, because it fundamentally challenges the validity of these findings by positing that treatment-dependent motion reduction is a primary driver of reported signal changes. We employ a novel slice-wise motion correction algorithm to decouple true neural activity from motion artifacts, a method significantly more sensitive than standard volume-based techniques. This new and substantively different approach is expected to allow us to overcome the current limitations of inconsistent cross-study reproducibility and potential false-positive biomarker identification caused by uncorrected between- and within-volume movement. Specifically, the current study adds several innovative components worth highlighting:

- We will address the association of ADHD treatment effects with changes in head motion during imaging, which is a significant problem that has been neither quantified nor managed. Both the known therapeutic effects of medication and our preliminary data indicate that this is a major confound in the literature that has yet to be identified, studied, or addressed.
- We will quantify how significant and variable the issue of head motion is when evaluating stimulant phMRI in ADHD. This is currently unknown and is a critical factor in interpreting and comparing findings in existing literature. It is also critical for guiding the design of future studies aiming, for example, to dissociate changes in movement with changes in overall treatment response.
- We will use our novel, advanced motion correction algorithms (slice-wise rather than volume-wise) that account for within-volume as well as between-volume movement, resulting in substantially higher quality images that reflect biological findings, rather than artifacts. This algorithm not only corrects with superior precision compared to standard processing streams but can also more accurately estimate the amount of movement over an fMRI run, allowing for better estimation of and removal of motion as a covariate. Our team uniquely focuses on the methodological aspects of motion correction as a prerequisite to investigating clinical impacts of treatment. Thus, we are optimally positioned to address this unique and important question.
- This project will pave the way to identifying accurate and reproducible regions that can be targeted with alternative treatments with fewer side effects than stimulants. This long-term goal is aimed at improving the quality of life and life/school/work performance of those whose lives are affected by ADHD.

## C. APPROACH

**C1. Patient population.** This study will include the collection and analysis of prospectively acquired neuroimaging and behavioral data from 150 typically developing participants between 12 and 18 years of age with ADHD who have been established on (dex)methylphenidate treatment for at least 3 months, with good efficacy, and are able to follow verbal and written directions in their native language. This sample size was selected with the expectation that 120 of these individuals will both (1) complete the study and (2) produce usable data. Participants will be recruited from the general community, the PI's clinical practice at BCH (about 100 unique ADHD patients seen annually), and from the Translational Neuroscience Center's Research Participant Registry (RPR), which contains over 30,000 potential local research participants.

**C2. Study design.** The proposed studies will be conducted using a repeated measures design, with four sessions per participant, each lasting 90 minutes. Importantly, each ADHD participant will serve as his/her own control; the design involves four scanning sessions that are identical, except two will be on-stimulant and the other two off-stimulant. Scan session order for on vs. off will be pseudorandomized across participants and across sessions within participant. Scan sequence order within-session will also be pseudorandomized. We will employ a multi-modal fMRI battery designed to elicit distinct cognitive states, provoke variable head motion profiles, and isolate the neural circuitry of attention and inhibition. Each session will include one T1-weighted scan, one T2-weighted scan, four task fMRI runs, and four functional connectivity (fc) fMRI runs; details for each scan type/task are described below with the corresponding Aim.

**Imaging acquisition and slice prescription.** All neuroimaging data will be acquired on a Siemens 3T Prisma MRI scanner using a 128-channel phased-array head coil. The acquisition protocol is derived from the harmonized Adolescent Brain Cognitive Development (ABCD) Study protocol to ensure high-resolution, multi-modal compatibility and reproducibility.<sup>37</sup> During each session, we will acquire structural images and fMRI images.

**Structural images** will include a **high-resolution, 3D, T1-weighted (T1w)** magnetization-prepared rapid gradient-echo (MPRAGE) sequence for anatomical segmentation and registration. Parameters will be: Voxel size

=  $1.0 \times 1.0 \times 1.0 \text{ mm}^3$  isotropic; TR = 2500 ms; TE = 2.88 ms; Inversion Time (TI) = 1060 ms; Flip Angle =  $8^\circ$ ; FOV =  $256 \times 256 \text{ mm}$ ; and Acceleration factor (iPAT) = 2. We will also acquire **high-resolution T2-weighted (T2w)** sampling perfection images with application-optimized contrasts using different flip angle evolution (SPACE) sequence to assist in surface reconstruction. Parameters for this sequence will be: Voxel size =  $1.0 \times 1.0 \times 1.0 \text{ mm}^3$  isotropic; TR = 3200 ms; TE = 563 ms; and FOV =  $256 \times 256 \text{ mm}$ . Functional imaging will be acquired for task and resting state imaging. For **task phfMRI (BOLD)**, we will use a T2\*-weighted multi-band (simultaneous multi-slice) echo-planar imaging (EPI) sequence (modified-Center for Magnetic Resonance Research (CMRR) sequence).<sup>41</sup> This sequence provides high temporal resolution to support the proposed slice-based motion correction analysis. Parameters will be: Voxel size =  $2.4 \times 2.4 \times 2.4 \text{ mm}^3$  isotropic; TR = 1060 ms; TE = 30 ms; Flip Angle =  $52^\circ$ ; Multi-band Acceleration Factor = 4; FOV =  $216 \times 216 \text{ mm}$ ; 60 slices (no gap). **Distortion correction** will be performed via fieldmap-less Synthetic Distortion Correction (SDC) to correct for geometric distortions (susceptibility artifacts).<sup>42</sup>

**C3. Preliminary data on in-scanner motion and motion correction in the ADHD population.** Our preliminary data running phMRI imaging studies in 10 adolescents with ADHD have demonstrated a very broad range of head movement across participants (**Fig. 2**). Notably, off-stimulant, there appeared to be high-motion participants and low-motion participants. On-stimulant, the range of head motion was smaller, and high-motion participants uniformly showed reductions in head motion, while low-motion participants showed similar motion, and a handful showed slightly greater motion. Thus, it is clear that motion-related changes are vastly different across on-/off-stimulant conditions for high-motion participants relative to low-motion participants with ADHD.

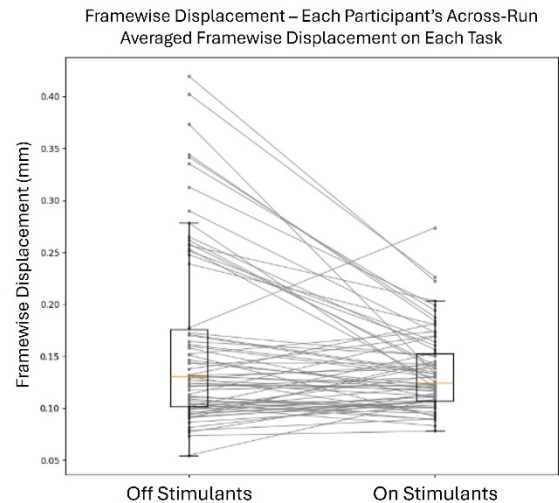
**C4 Aim 1: To study the effects of superior motion correction on outcomes of within-subject task phMRI in stimulant-responsive ADHD by comparing standard and advanced (slice-by-slice) motion correction.**

**Introduction.** We have an incomplete understanding of the mechanisms by which stimulants improve ADHD symptoms. This is, in part, due to the variability of outcomes across phMRI studies. The effects of treatment on hyperactivity lead to differences in signal-to-noise across on/off phMRI conditions and may be a factor in study outcome differences. This is a significant confound in ADHD phMRI studies that has not been addressed previously. The objective of this aim is to evaluate how phMRI outcomes differ in individuals with ADHD on- vs. off-stimulant when comparing volume- (current standard) vs. slice-wise (recently developed) motion correction. To attain this objective, we will test the working hypothesis that (1) study results, including regions identified and direction of signal change, are contaminated by motion artifacts in the pediatric ADHD population, (2) motion artifacts are not the same off- vs. on-stimulant in this population, and (3) that (1) and (2) contribute to both false-positive and false-negative results when running phMRI studies in this population. Our approach to test the working hypothesis will be to apply our innovative slice-based motion correction to more precisely (and therefore also more accurately) process images from stimulant phMRI in pediatric ADHD. The rationale for this aim is that its successful completion will both identify the amount of error introduced by motion in ADHD phMRI and illustrate a more accurate picture of the regions that differ with stimulant treatment in ADHD. Upon completion of Aim 1, it is our expectation that we will illustrate critical and profound effects of stimulants on head motion during imaging and illustrate how outcomes differ when motion is accounted for in a more precise manner. Such findings would be critically important because they are the first step toward our overall goal to improve outcomes in ADHD by better understanding the modulation of brain networks by stimulants.

**Preliminary Data for Aim 1**

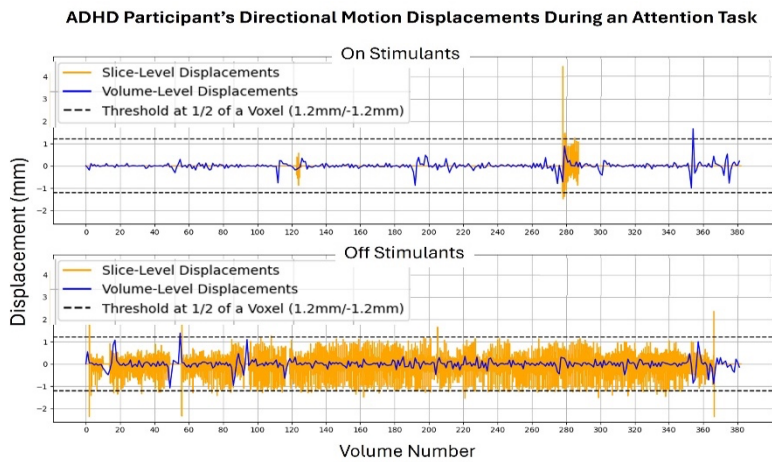
Stimulants reduce in-scanner motion during task phMRI, but not in a consistent fashion. Our preliminary data from pilot phMRI studies show clear evidence that many, but not all, individuals with ADHD show greater in-scanner head motion off-stimulant than on-stimulant (see **Fig. 2** above).

In-scanner motion is incompletely captured in task phMRI using current volume-by-volume measurements. Our team recently developed slice-by-slice motion characterization as a step towards superior alignment of MRI images. As discussed in the Significance section, for our current cohort, we applied this existing software to 10 individuals with ADHD participating in task phMRI studies on- vs. off-stimulant and compared movement



**Figure 2.** A significant proportion of participants with ADHD show far more head motion, as measured by average frame-wise displacement, while being scanned off stimulants than on stimulants.





**Figure 3.** Comparison of on-stimulant vs. off-stimulant scanning (top graph vs. bottom graph) and frame-wise vs. slice-wise computed head motion (blue vs. orange) in a pediatric ADHD participant. Motion displacement is shown with directional information (negative=left; positive=right) such that it is clear that there is within-volume back-and-forth jitter across slice groups that cancels out in the volume-wise motion measures.

**Task fMRI phMRI study design.** As outlined in **C2. Study Design** above, we will conduct task phMRI in 150 individuals with ADHD who are currently and effectively treated with stimulants, with the expectation that 120 of these individuals will both (1) complete the study and (2) produce usable data. Each participant will be scanned in four temporally distinct sessions, separated by at least 2 days but not more than 6 weeks, with two sessions on-stimulant and the other two off-stimulant. Each session will include identical sequences/tasks across sessions, and sequence order will be counter-balanced and pseudorandomized across session. The order of session acquisition (on- vs. off-stimulant) will be pseudorandomized such that half the subjects will be scanned off-stimulant first and the other half on-stimulant first. Study investigators will not be blinded to medication status since participants are taking or withholding their own clinically prescribed medications and providing a placebo is not feasible – given the strong psychoactive properties and high side-effect profile of stimulants, both participants and observers are typically aware whether a participant is on or off medication significantly above chance.

**Task conditions included for Aim 1.** The task battery will consist of two distinct acquisition types: (1) the Multi-Source Interference Task (MSIT) and (2) the Stop Signal Task (SST). To ensure sufficient signal-to-noise ratio (SNR) and reproducibility, we will acquire two runs of each paradigm, with each run lasting approximately 6 minutes. This yields a total task-related functional acquisition time of ~24 minutes per session.

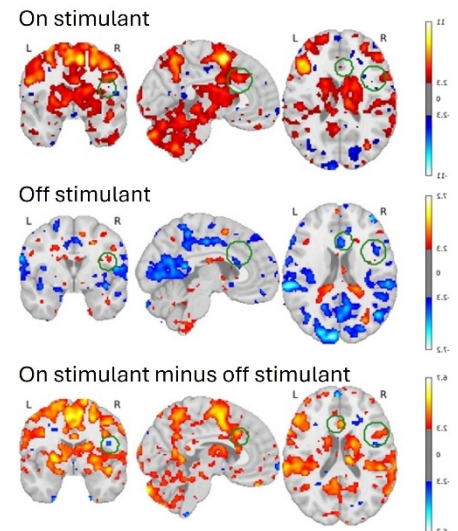
**The Multi-Source Interference Task (MSIT)** consistently and maximally activates and isolates **dACC**, even at the individual level. The task will be administered using a block design. The task involves presentation of three digits (e.g., "1 0 0", "2 2 1") and requires the participant to identify the unique number by indicating its position. In the **Control condition**, the target number's identity matches its position (e.g., "1 0 0", target is 1, position is 1). In contrast, the **Interference condition** positions the unique number in a location incongruent with its order (e.g., "2 1 2", target is 1, position is 2) and is flanked by response-incongruent distractors that are not in their order position and are not "zero." This task imposes a high cognitive load, generating "cognitive strain" that is sensitive to stimulant effects and typically associated with task-correlated motion. The activation maps shown in **Fig. 4** are representative maps for this task.

**The Stop Signal Task (SST)** will be used to activate and isolate the rIFG and measure response inhibition/action cancellation. The SST has been shown to be more right-lateralized and produce more robust rIFG activation than a typical go/no-go paradigm. This task uses an event-related design with a staircase tracking algorithm that dynamically adjusts to ensure each participant achieves a ~50% successful inhibition rate. This

using frame-wise displacement (FD) vs. slice group displacement (SGD) motion correction approaches. The term "frame-wise displacement" (FD) is used to refer to the motion measurements between sequential volumes of data, analogous to sequential frames of a movie reel. We found that (1) frame-wise and slice-wise displacements did not always take place in the same volume (**Fig. 3**) and (2) SGD occurred inconsistently across runs, as well as across slices in a given volume (**Fig. 3**).

**Stimulants affect task-related activity in the ACC and/or the rIFG.** We have also verified that task phMRI shows differences in the dACC and rIFG depending on whether the participant was on- or off-stimulant (**Fig. 4**). In particular, we have shown differences in dACC, in this case at the individual level, with greater activation on-stimulant.

Individual ADHD participant z-maps for the MSIT task on- versus off-stimulant



**Figure 4.** Z maps for an individual with ADHD who shows differences in dACC and rIFG activation during the multi-source interference task (MSIT) on- vs. off-stimulant. Color bars indicate scale for z maps. Green circles highlight the dACC and rIFG.

equates task difficulty across "on" and "off" medication sessions, ensuring that BOLD differences reflect neural efficiency rather than effort or performance disparities. **This task involves performing** a primary "Go" task (responding to left/right arrows), which is occasionally interrupted by a "Stop" signal (auditory tone or visual cue) appearing after a variable Stop Signal Delay (SSD), which means that participants must stop an already ongoing task rather than deciding whether to stop or go each trial. The SST has been established as a task that specifically recruits the rIFG during "Stop" trials. Importantly, the sudden requirement to withhold a motor response frequently provokes distinct "jerk" head movements, providing a critical testbed for our slice-wise motion correction algorithm.

**Run Order and Counterbalancing.** To minimize fatigue and order effects, the presentation order of the four task types will be counterbalanced across participants and across the two sessions (on/off medication). Behavioral responses (reaction time and accuracy) will be recorded for all active tasks.

**Motion correction methods.** In the current study, we will use (1) standard, volume-based motion correction via MCFLIRT from the FMRIB Software Library (FSL) package and (2) retrospective, slice-by-slice motion correction methods using custom Python scripts. Both methods will be applied to each raw dataset independently for direct comparison of signal fidelity and artifact reduction.

**Volume-based motion correction:** We will utilize the standard **MCFLIRT** tool (Motion Correction FMRIB's Linear Image Registration Tool) from the FSL software library. This method assumes the head remains stationary during the acquisition of a single TR (volume). For each run, all volumes will be registered to a single reference volume (e.g., the middle volume of the time series) using a rigid-body transformation (6 degrees of freedom: 3 translations, 3 rotations). Resampling will be performed using trilinear interpolation to generate the final motion-corrected 4D file. Standard slice-timing correction will be applied, which also presumes no motion occurred during each volume acquisition.

**Slice-based motion correction:** We will employ a custom Python-based pipeline designed to correct at the acquisition level (slice group) rather than the volume level, measuring and mitigating "within-volume" motion artifacts. This approach is particularly critical for multiband acquisitions where rapid head movements can cause misalignment between slice groups within a single TR. Correction will be achieved using the following steps: **(1) Estimation:** Rigid-body translation and rotation parameters will be calculated for each simultaneously acquired slice group (multiband factor = 4) relative to the reference geometry. **(2) Reconstruction:** Using precise slice-timing information, the data from each slice group will be moved to its spatially corrected location within a **temporal super-resolution matrix**. **(3) Correction:** We will utilize **sparse spline interpolation** to perform slice-timing correction and motion realignment simultaneously. *This single-step resampling reconstruction generates a new, motion-stabilized volume-by-volume 4D file while minimizing the blurring effects of multiple interpolation stages.*

## **Hypothesis-Driven Analyses**

**Primary outcome:** We hypothesize that there will be greater in-scanner head movement in off- compared to on-stimulant in ADHD participants that impacts identified regions of difference. We will test this hypothesis by comparing task phMRI outcomes on- vs. off-stimulant using standard volume-based correction vs. our novel slice-based correction. *We expect that* accounting for movement differences on- vs. off-treatment using higher-precision, slice-based motion correction will lead to different regional task-related phMRI activation/deactivations than when using volume-based motion correction.

**Power analysis:** Our recruitment target of **N=150** participants is designed to yield at least **N=120** fully evaluable individual datasets, after a conservative ~20% attrition rate due to non-compliance, technical failure, or excessive motion. This sample size is specifically chosen not merely to detect main effects but to capture the full population range of baseline motion and the heterogeneous magnitude of motion reduction induced by stimulants. With N=120 in a within-subject design (on- vs. off-stimulant), we will have >99% power to detect the moderate effect sizes typically reported in ADHD phMRI literature (Cohen's  $d = 0.5$ ) at  $\alpha = 0.05$ . Crucially, this larger sample size drastically improves our sensitivity to small effects, allowing us to reliably detect effect sizes as small as Cohen's  $d = 0.26$  with 80% power. This sensitivity is critical for identifying subtle, regionally specific activation changes in subcortical structures (Aim 1) or weak network shifts (Aim 2) that may have been missed in underpowered previous studies (N=20-40). The primary methodological goal is to validate Slice-Based Motion Correction (SBMC) against Standard Volume-Based Correction (VBMC). An N=120 sample allows for robust interaction analyses (Method X Treatment) and ensures we can additionally test the hypothesis that slice correction has a larger effect/benefit in the most hyperactive (and thus most vulnerable to motion artifacts) segment of the ADHD population, characterizing the *range* of motion effects rather than just the mean.



## Statistical Analyses

Preprocessing Pipeline: All MRI data will be converted to the BIDS standard and processed using a custom pipeline integrating FSL and Nilearn/Python tools. Structural T1w images will be skull-stripped and registered to the MNI152 template using linear registration (FLIRT) and non-linear (FNIRT) registration steps. Functional data (Task and Resting State) will undergo two parallel processing streams for comparison: (1) **Standard Stream:** Volume-based motion correction (FSL MCFLIRT), slice-timing correction, and spatial smoothing (4mm FWHM), and (2) **Advanced Stream:** Our custom slice-based motion correction (simultaneous motion and slice-timing correction via sparse interpolation), followed by spatial smoothing (4mm FWHM Gaussian kernel). Both streams will undergo susceptibility distortion correction using topup (FSL) or Synth SDC, and then will be co-registered to the native space T1w structural image. Nuisance regression will be performed using the General Linear Model (GLM) module in Nilearn to remove effects stemming from white matter (WM), cerebrospinal fluid (CSF), motion parameters (6 rigid-body + temporal derivatives), and global signal (to rigorously control for systemic physiological noise) for task or functional connectivity analyses per standard practice.

First-Level (Subject) Analysis: For each participant and session (on/off-stimulant), GLMs will be constructed using Nilearn and tested via permutation-based FSL Permutation Analysis of Linear Models (PALM) software. **MSIT:** Regressors for Interference and Control blocks will be modeled. The primary contrast of interest is Interference condition vs. Control condition, with a focus on regions showing greater activation during the Interference condition. **SST:** Regressors for Successful Stop, Failed Stop, and Go trials will be modeled. The primary contrast is Successful Stop vs. Go, with a focus on regions showing greater activation during Successful Stop frames/volumes than Go frames/volumes. All regressors will be convolved with a canonical hemodynamic response function (HRF).

Group-Level & Methodological Comparison: To rigorously test Aim 1, we will employ a Linear Mixed Effects (LME) model (implemented via Python statsmodels or R lme4) that handles the hierarchical nature of the data (Sessions nested within Subjects). Our primary analysis will use a region of interest (ROI) approach for our *a priori* regions of interest (dACC for MSIT; rIFG for SST). We will extract mean beta values and residual variance from each ROI and compare these metrics using a 2x2 factorial design consisting of Treatment (on vs. off) and Correction Method (slice vs. volume).

Hypothesis Testing: We will construct a GLM with factors for treatment (on vs. off) and correction method (slice vs. volume). Covariates will include Age, Sex, and Mean Frame-wise Displacement (FD) or Slice-wise Displacement (SD). Using a permutation-based paired t-test of variance, we will test for a main effect of method, that is, does slice correction significantly reduce residual error variance compared to volume correction? Using a mixed model ANOVA, we will also test for interaction effects, that is, does the detection of the treatment effect (on vs. off) change significantly depending on the correction method used? To test if benefits are driven by high-motion subjects, we will include a regressor for "Baseline Motion" to assess if the divergence between volume and slice correction scales with the amount of subject movement.

Motion Characterization: We will calculate displacement traces for both Frame-wise Displacement (FD) and Slice Group Displacement (SGD) alignments as described above. These traces and their 1<sup>st</sup> derivatives will be considered for use as additional regressors in the individual-level GLMs to remove remaining motion effects.

Correction for Multiple Comparisons: Voxel-wise group maps will be thresholded using permutation-based family-wise error and false-discovery-rate approaches ( $p < 0.05$ , corrected) to identify whole-brain differences in activation patterns between the two correction methods. ROI-based analyses will be corrected for number of simultaneous tests/contrasts performed per FSL PALM standard practice.

**Potential Problems and Alternative Approaches:** Our working hypothesis is that greater head motion in ADHD participants off-stimulant than on-stimulant will lead to reduced fidelity of off-stimulant images due to motion artifact, and that standard volume-based motion correction will not address this sufficiently or as well as slice-based motion correction. Although our preliminary data and previous literature showing significant inconsistencies in phMRI of ADHD psychostimulant treatment support the hypothesis for this Aim, there is a remote possibility that it could be invalidated when tested objectively. For example, head motion may be more subtle than stimulants can correct or slice-wise. Below, we outline potential challenges that may arise and the steps we would take to address them.

Slice correction does not fully salvage high-motion data. While slice-wise correction reduces artifacts by correcting for intra-volume misalignment, it cannot correct for "spin history" effects (excitation of the wrong slice due to movement). If residual artifacts persist, we will implement "scrubbing" (censoring) of TRs where slice-based displacement exceeds 0.5 mm. We will then compare the number of surviving TRs between methods.

We hypothesize that slice correction will require significantly *less* scrubbing than volume correction to achieve comparable temporal SNR (tSNR).

No significant "On vs. Off" treatment effect is observed, regardless of correction method. If we fail to replicate the canonical dACC/rIFG activation changes even with superior motion correction, this would be a significant null finding, suggesting that previous literature findings may have been driven largely by motion artifacts (false positives). In this case, our primary outcome would shift to characterizing the *nature* of the artifact itself. We would perform a "null data" simulation, adding simulated motion to clean data (from identified low-motion participants) to demonstrate how volume-based correction creates spurious activation patterns that slice-based correction removes.

Over-correction or Smoothing. There is a risk that aggressive interpolation could smooth biological signal. To mitigate this, we use sparse interpolation in our algorithm to minimize data manipulation. We will validate signal integrity by calculating tSNR maps for both methods; we expect tSNR to increase in grey matter with slice correction without the blurring characteristic of excessive smoothing.

Future Direction. Even if slice-based correction does not recover "lost" signal in all subjects, the ability to *quantify* the exact amount of within-volume motion (typically invisible to researchers) is a major contribution. This project will establish a new quality control metric, **Slice-Based Displacement (SBD)**, that future studies can use as a covariate of no interest, fundamentally improving the rigor of neuroimaging in high-motion pediatric populations.

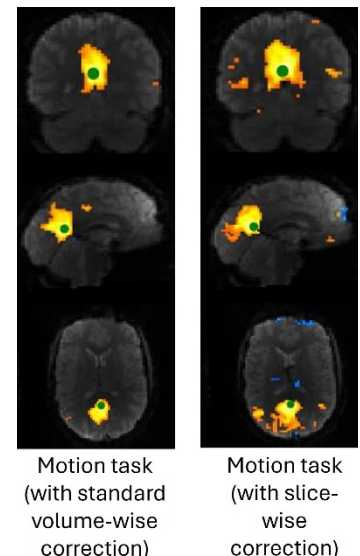
While slice-wise motion correction substantially reduces motion artifacts (by an order of magnitude proportional to slice number), within-slice motion can still potentially blur the raw data. While this itself cannot be corrected, the elegance of our ability to quantify motion here highlights that future studies can characterize just how much participants moved. What this means is that studies can quantify just how much their participants moved, leading to new meta-analysis variables of no interest (or interest) that can be taken into account when interpreting differences across studies. The ability to do this is, in and of itself, an important contribution of this work.

## **C5 Aim 2: To study the effects of superior motion correction on outcomes of within-subject on/off-stimulant rs-fcMRI (connectivity) by comparing standard vs advanced (slice-by-slice) motion correction**

**Introduction.** The objective of this aim is to determine and account for the effects of head motion in individuals with ADHD on assessing stimulant treatment influence on functional connectivity, and to account for these effects using superior motion correction algorithms. To attain this objective, we will test the working hypothesis that in-scanner head motion impairs our ability to accurately assess resting state network connectivity changes to treatment. In particular, it has been shown that regions proximal to a seed region are more impacted by head motion than regions further away; as a result, overall network connectivity (as opposed to specific regions) would thus be substantially impacted by motion, since connectivity across a network must be in sync to show increased connectivity. Here, we implement superior motion correction at the slice rather than full volume level, with the goal of minimizing these network effects. Our approach to test the working hypothesis will be to compute functional connectivity measures with both approaches and test for improvement in network detection or reduction in variation in the signal. The rationale for this aim is that its successful completion will both identify the amount of error introduced by motion in ADHD rs-phMRI and illustrate a more accurate picture of the connections that are affected by stimulant treatment in ADHD. Upon completion of Aim 1, it is our expectation that we will illustrate critical and profound effects of stimulants on head motion during resting state imaging and illustrate how the appearance of networks differs when motion is accounted for in a more precise manner. Such findings would be critically important because they demonstrate the potential for compensating for motion and allowing for additional data to be used on a per-participant basis compared to aggressive scrubbing.

### **Preliminary Data for Aim 2**

In-scanner motion is also observed during functional connectivity fMRI. Our preliminary data from a recent publication (**Fig. 5**) evaluating effects of slice-wise motion scrubbing on functional connectivity fMRI confirmed that head motion is a



**Figure 5.** Comparison of posterior cingulate cortex (PCC) seed-based functional connectivity map differences using volume-wise vs. slice-wise motion correction from our recent publication.<sup>2</sup> Data are from a "medium motion" rs-fMRI run in a participant with ADHD.

factor in this imaging modality, even in the absence of tasks with motor responses.<sup>3</sup> Moreover, slice-wise correction, along with low-rank completion/sparse interpolation methods, has shown proof of concept that connectivity maps appear different depending on analysis method.

In-scanner motion is incompletely captured in rs-fMRI phMRI using current volume-by-volume measurements. Our team recently developed slice-by-slice motion characterization to achieve more accurate alignment of MRI images. For our current cohort, we applied this existing software to 10 individuals with ADHD participating in rs-fMRI phMRI studies on- vs. off-stimulant and compared movement using frame-wise displacement (FD) vs. slice group displacement motion correction approaches. We found (Fig. 6) that (1) frame-wise and slice-wise displacements did not always take place in the same volume and (2) SGD occurred inconsistently across runs, as well as across slices in a given frame/volume.

**rs-fMRI phMRI Study Design.** As outlined in **C2. Study Design** above, we will conduct task phMRI in 150 individuals with ADHD who are currently and effectively treated with stimulants, with the expectation that 120 of these individuals will both (1) complete the study and (2) produce usable data. Each participant will be scanned in four temporally distinct sessions, separated by at least 2 days but not more than 6 weeks, with two sessions on-stimulant and the other two off-stimulant. Each session will include identical sequences/tasks across sessions, and sequence order will be counterbalanced and pseudorandomized across session. The order of session acquisition (on- vs. off-stimulant) will be pseudorandomized such that half the subjects will be scanned off-stimulant first and the other half on-stimulant first. Study investigators will not be blinded to medication status since participants are taking or withholding their own clinically prescribed medications and providing a placebo is not feasible – given the strong psychoactive properties and high side-effect profile of stimulants, both participants and observers are typically aware whether a participant is on or off medication significantly above chance.

**Functional Connectivity Conditions Included for Aim 2.** The functional connectivity battery will consist of two distinct acquisition types: (1) Resting State, and (2) Naturalistic Movie Watching. To ensure sufficient SNR and reproducibility, we will acquire two runs of each functional connectivity paradigm, with each run lasting approximately 6 minutes. This yields a total functional connectivity-related acquisition time of ~24 minutes per session.

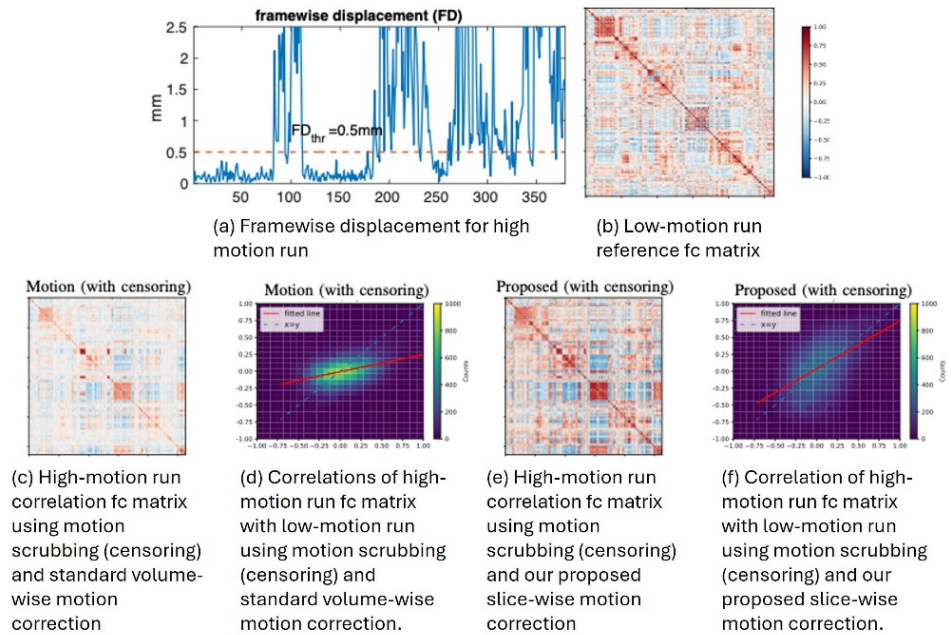
Resting state: For resting state runs, participants will be instructed to relax and try to stay awake while fixating on a small white cross embedded in a dark background.

Naturalistic movie watching: The two runs of movie watching will consist of (1) viewing of a 10-minute clip from the movie *Despicable Me* (01:02:09–01:12:09)<sup>43</sup> and (2) viewing the "Inscapes" animation<sup>44</sup> a low-load, abstract, non-social animation designed to facilitate high-quality resting-state acquisition in pediatric populations. "Inscapes" has been shown to significantly reduce head motion and maintain wakefulness compared to standard fixation, while yielding functional connectivity matrices highly comparable to traditional resting-state methods. This approach is critical for minimizing attrition due to motion in our adolescent ADHD cohort.

**Motion Correction Methods.** We will use motion correction methods identical to those described for Aim 1.

### Hypothesis-Driven Analyses

Primary outcome: We hypothesize that standard volume-based motion correction fails to account for the complex, high-frequency head motion characteristic of ADHD, leading to spurious differences in functional



**Figure 6.** Illustration of the superior effects of slice-wise motion correction, compared to standard volume-wise correction, on functional connectivity fMRI data from the ABCD dataset from our recent publication.<sup>2</sup>

connectivity that are currently misattributed to treatment effects. We will test this hypothesis by comparing rs-fMRI phMRI functional connectivity on- vs. off-stimulant using standard volume-based correction vs. our novel slice-based correction. We expect that some of the "standard" on- vs. off-stimulant connectivity differences will be significantly attenuated when applying our novel slice-based correction, suggesting many prior findings are motion artifacts, while true neural differences will be more easily identifiable once motion is properly corrected.

Power analysis: The sample size calculation described in Aim 1 (150 recruited; 120 evaluable) applies equally to Aim 2. Resting-state functional connectivity studies of two conditions typically achieve stability with 40-50 participants. Our target of 120 provides >95% power to detect moderate correlations ( $r=0.3$ ) and allows us to detect interaction effects (Method X Treatment) within the resting-state data, even after correcting for multiple comparisons across a 264x264 ROI matrix.

### **Statistical Analyses**

Preprocessing Pipeline: We will utilize a hybrid pipeline that integrates our novel motion correction steps with the established the Computational Brain Imaging Group (CBIG) fMRI Preprocessing Pipeline (Thomas Yeo Lab, Singapore/Harvard) to ensure rigorous standardized handling of resting-state data. First, raw fMRI time series will undergo either (A) volume-based correction (FSL MCFLIRT) or (B) slice-based correction (our custom algorithm) as described in Aim 1.

Functional Connectivity Preprocessing: Following the initial motion correction, both datasets will enter the CBIG pipeline (<https://github.com/ThomasYeoLab/CBIG>). Steps include: **1) Nuisance Regression** of 18 parameters: 6 motion parameters, their temporal derivatives, and the first principal components of white matter and CSF signals (CompCor). Global Signal Regression (GSR) will be applied to remove widespread physiological noise, which is critical for unmasking specific network anticorrelations; **2) Censoring** of volumes with excessive remaining motion ( $FD > 0.2\text{mm}$ ) to ensure rigorous comparison. We hypothesize fewer volumes will require censoring in the slice-based stream; **3) Bandpass filtering** ( $0.009\text{Hz} < f < 0.08\text{Hz}$ ) to isolate low-frequency fluctuations; and **4) spatial smoothing** with a 4 mm FWHM Gaussian kernel.

Motion Characterization: We will calculate displacement traces for both Volume Displacement (FD) and Slice Group Displacement (SGD) alignments. We will quantify "missed motion" by comparing the Area Under the Curve (AUC) of FD vs. SGD, testing the hypothesis that standard volume-based metrics significantly underestimate true head motion in ADHD.

Individual-Level Statistical Analyses: For each participant, we will generate: **1) A region–region functional connectivity matrix** using an available set of 264 identified brain regions of interest (ROIs).<sup>45</sup> Each ROI will be modeled as a sphere of radius 6 mm and a correlation matrix will be derived from the Pearson product moment correlations between the time series extracted from each pair of ROIs, and **2) Fisher-z-transformed correlation maps** between our *a priori* seeds (dACC and rIFG) and all other voxels.

Group-Level Statistical Analyses & Methodological Comparison: All group-level statistical modeling will be conducted using FSL PALM, which allows for rigorous non-parametric testing and family-wise error (FWE) and false discovery rate (FDR) correction while accounting for the complex exchangeability blocks inherent in our within-subject design.

Hypothesis Testing: We will construct a GLM with factors for treatment (on vs. off) and correction method (slice vs. volume). Covariates will include Age, Sex, and Mean Frame-wise Displacement (FD) or Slice-wise Displacement (SD). Using a permutation-based paired t-test of variance, we will test for a main effect of stimulant, controlling for motion and a main effect of correction method. We will then use repeated-measures ANOVAs in PALM to test our central hypothesis: that the *perceived* difference between off and on states is significantly distinct depending on the motion correction applied. To test if benefits are driven by high-motion subjects, we will include a regressor for "Baseline Motion" to assess if the divergence between Volume and Slice correction scales with the amount of subject movement.

Correction for Multiple Comparisons: Voxel-wise group maps will be thresholded using permutation-based family-wise error and false-discovery-rate approaches ( $p < 0.05$ , corrected) to identify whole-brain differences across the whole brain (seed-based) or the whole matrix (network analysis). Statistical significance in fcMRI maps will also be assessed using Threshold-Free Cluster Enhancement (TFCE) within PALM, corrected for family-wise error (FWE) at  $p < 0.05$ . ROI-based analyses will be corrected for number of simultaneous tests/contrasts performed per FSL PALM standard practice.

### **Potential Problems and Alternative Approaches**

**Data Quantity and SNR Disparities:** A primary challenge in ADHD rs-fMRI is excessive motion "Off-stimulant", potentially leading to significantly less usable data compared to "On-stimulant". While our sample size (N = 120 evaluable) is powered to absorb attrition, we will rigorously control for SNR differences by employing *data truncation*, ensuring the "On" session duration is cropped to match the usable duration of the "Off" session for each participant. We further hypothesize that our slice-based correction will salvage volumes that would otherwise be discarded by standard censoring (FD > 0.2 mm), maximizing data retention.

**Signal Artifacts from Correction:** There is a risk that the interpolation inherent in slice-based correction could introduce temporal smoothing or spectral shifts. To mitigate this, we will *compare tSNR* maps and *power spectral density plots* between pipelines. If spectral shifts are detected, we will adjust the bandpass filter limits in the CBIG pipeline to align the spectral profiles. We will also validate findings using parallel preprocessing streams (Global Signal Regression vs. CompCor) to ensure robust noise removal.

**Heterogeneity Masking Effects:** If biological heterogeneity in ADHD masks group-level changes in canonical network, we will pivot to *individual-specific network mapping* (e.g., <sup>46,47</sup>). This allows us to assess changes in topological organization (network integration vs. segregation) closer to the individual anatomy rather than relying solely on group-averaged seed maps.

**Null Interaction:** If we find no significant difference between correction methods, we will pivot to characterizing the *motion-FC relationship*. We will perform simulations adding realistic motion profiles to clean data to mathematically quantify the specific "tipping point" at which volume-based correction fails, providing a critical quality control benchmark for the field.

## C6. Summary Addressing NIH Guidelines on Scientific Rigor

**Consideration of sex and other biological variables:** All analyses will include sex and age as covariates. We will conduct sub-analyses to directly test whether our results hold true equally in males and females.

**Transparency/Reproducibility:** Our imaging data and motion correction pipelines will be shared on the "open science" OpenNeuro or Harvard Dataverse platforms per the attached Data Management Plan and freely distributed. More details can be found in the attached Data Management Plan.

## C7. Timetable and Benchmarks for Success

|                                                                                                           | Year 1 |    |    |    | Year 2 |    |    |    | Year 3 |    |    |    | Year 4 |    |    |    | Year 5 |    |    |    |
|-----------------------------------------------------------------------------------------------------------|--------|----|----|----|--------|----|----|----|--------|----|----|----|--------|----|----|----|--------|----|----|----|
|                                                                                                           | Q1     | Q2 | Q3 | Q4 | Q1     | Q2 | Q3 | Q4 | Q1     | Q2 | Q3 | Q4 | Q1     | Q2 | Q3 | Q4 | Q1     | Q2 | Q3 | Q4 |
| Full Investigators Meeting                                                                                |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |
| Participant Recruitment and Scanning                                                                      |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |
| Quality control of neuroimaging data (Aims 1 and 2)                                                       |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |
| Quality control of phenotypic data (Aims 1 and 2)                                                         |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |
| Refinement of slice-based registration and sparse-matrix interpolation algorithms (Aims 1 and 2)          |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |
| Manuscript on containerized pipeline for slice-based registration and sparse-matrix interpolation of fMRI |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |
| Analysis of motion data off and on stimulant compared to out-of-scanner measures                          |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |
| Manuscript on the effect of stimulants on head motion, out-of-scanner measures, and time-in-scanner       |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |
| Analysis of task-based results off and on stimulant (Aim 1)                                               |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |
| Manuscript on the effect of stimulants during tasks with improved motion capture (Aim 1)                  |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |
| Analysis of functional connectivity results off and on stimulant (Aim 2)                                  |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |
| Manuscript on the effect of stimulants on functional connectivity with improved motion capture (Aim 2)    |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |
| Preparation for Data and Code sharing in open-source databases                                            |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |
| Manuscript on individualized localization of stimulant effects in ADHD (Aims 1 and 2, exploratory)        |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |        |    |    |    |

## C8. Future Directions

If successful, this project will define a critical variable associated with variability in findings across previous phMRI and rs-fMRI studies in individuals with ADHD. This directly suggests future studies should incorporate slice-wise motion correction, take motion into account as a covariate in imaging analyses, and, if possible, perform real-time monitoring to prevent acquisition of corrupted slices and volumes. This project will also generate important deliverables that directly lead into additional high-impact studies, including: 1) Studies that begin to parse biological variables (e.g., etiologic heterogeneity, side effects, etc.) once motion has been better accounted for and (2) To use target localization identified from the current study to begin fueling design of future TMS and neurofeedback studies. If successful, the proposed project will lead to reconciling differences between findings in prior studies and to understanding design and analysis features critical to design of future projects.



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